2026 06 04
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

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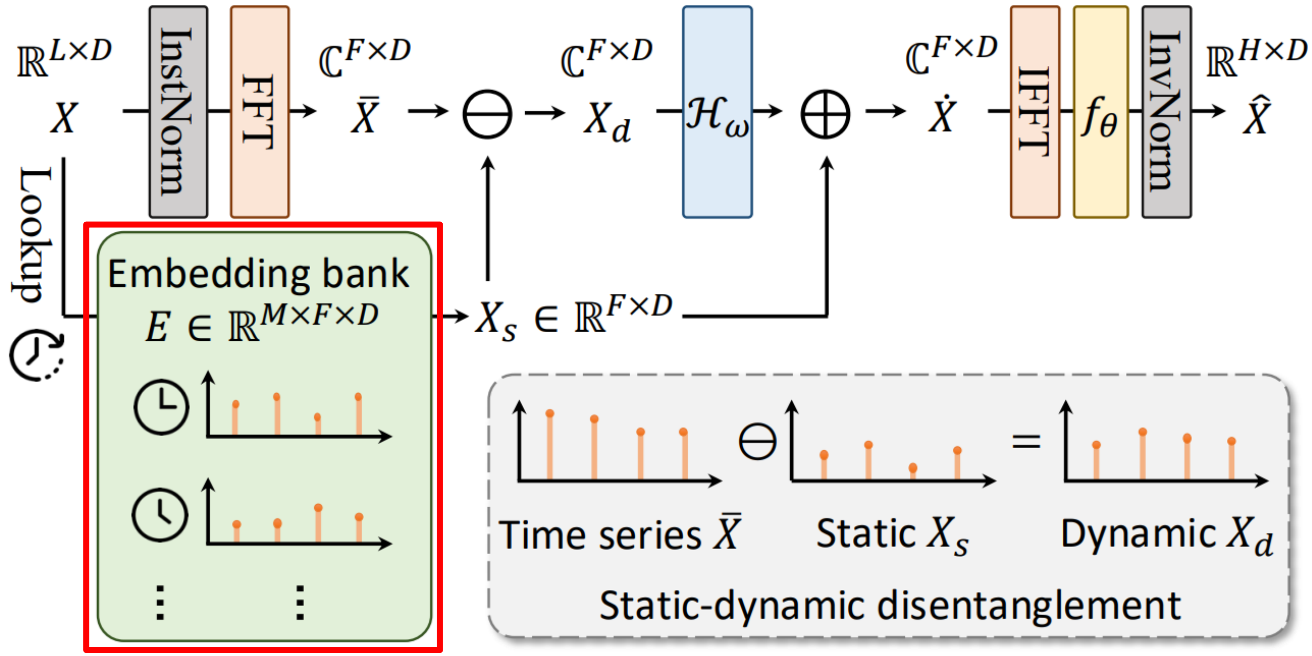
이번 주 진행사항

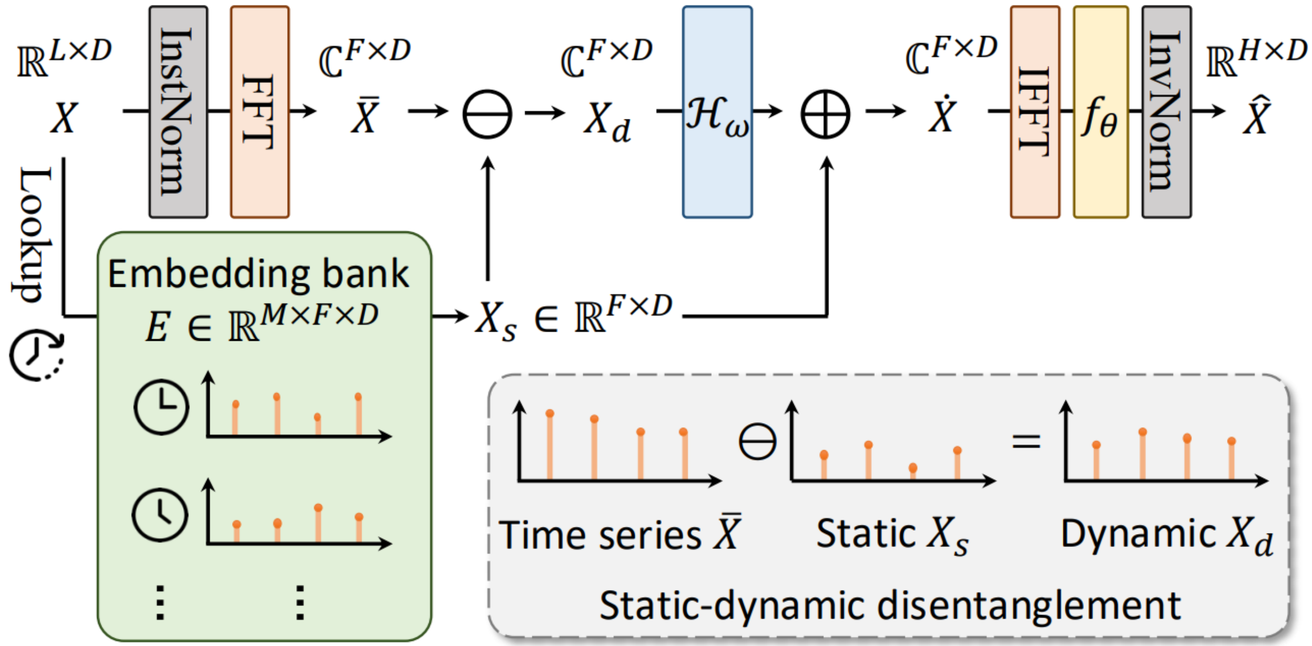
- TimeEmb
  - 아이디어
  - 재현 실험

Overview



- Embedding bank
  - 입력 데이터의 마지막 타임스탬프를 기반으로 해당 시간 슬롯의 Embedding을 Lookup
  - 동일 시간 슬롯의 모든 샘플이 하나의 Embedding을 공유하며, Backpropagation을 통해 반복적으로 나타나는 공통 주파수 패턴을 학습
  - 학습된 Embedding은 해당 시간 슬롯에서 지속적으로 관측되는 평균적인 패턴을 나타내며, 이를 정적 성분 X_s로 간주

아이디어



- X_s의 한계
  - 모든 공통적이고, 안정적인 패턴을 단일 정적 성분 X_s로 표현
  - 실제 시계열은 다양한 스케일의 패턴이 존재하기 때문에 X_s 로는 충분히 표현하기 어려움
  - 그렇기 때문에 일부 안정적인 패턴이 동적 성분 X_d에 남을 수 있음
- 아이디어
  - 해당 한계점을 보완하는 아이디어 구상

실험 세팅

- 사용한 모델: TimeEmb
- 재현 실험 데이터셋: ETTh1, ETTm1

Experiment	ETTh1
Learning rate	5×10^{-4}
Epoch	100
Batch size	128
Loss function	MSE Loss
Seq_len	96
Pred_len	96/192/336/720
Layers	3
Expert	4~16

재현 실험 (ETTh1)

	ETTh1 Paper		ETTh1 Reproduction	
Prediction length	MSE	MAE	MSE	MAE
96	0.366	0.387	0.366	0.387
192	0.417	0.416	0.417	0.416
336	0.457	0.436	0.456	0.436
720	0.459	0.460	0.459	0.459

재현 실험 (ETTm1)

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	ETTm1 Paper		ETTm1 Reproduction	
Prediction length	MSE	MAE	MSE	MAE
96	0.304	0.343	0.304	0.343
192	0.354	0.373	0.353	0.373
336	0.379	0.393	0.380	0.393
720	0.435	0.428	0.433	0.426

이후 계획

- TimeEmb
  - 아이디어 적용 후 실험

부록

Performance comparison

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Model		TimeEmb (ours)		CycleNet 2024		Fredformer 2024		FilterNet 2024		iTransformer 2024		PatchTST 2023		FITS 2024		FreTS 2023		DLinear 2023	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.366±0.001**	**0.387±0.001**	0.378	0.391	0.373	0.392	0.375	0.394	0.386	0.405	0.394	0.406	0.386	0.396	0.395	0.407	0.386	0.400
	192	**0.417±0.001**	**0.416±0.001**	0.426	0.419	0.433	0.420	0.436	0.422	0.441	0.436	0.440	0.435	0.436	0.423	0.448	0.440	0.437	0.432
	336	**0.457±0.001**	**0.436±0.001**	0.464	0.439	0.470	0.437	0.476	0.443	0.487	0.458	0.491	0.462	0.478	0.444	0.499	0.472	0.481	0.459
	720	**0.459±0.002**	0.460±0.001	0.461	0.460	0.467	**0.456**	0.474	0.469	0.503	0.491	0.487	0.479	0.502	0.495	0.558	0.532	0.519	0.516
	avg	**0.425±0.001**	**0.425±0.001**	0.432	0.427	0.435	0.426	0.440	0.432	0.454	0.447	0.453	0.446	0.451	0.440	0.475	0.463	0.456	0.452
ETTh2	96	**0.277±0.001**	**0.328±0.001**	0.285	0.335	0.293	0.342	0.292	0.343	0.297	0.349	0.288	0.340	0.295	0.350	0.309	0.364	0.333	0.387
	192	**0.356±0.001**	**0.379±0.001**	0.373	0.391	0.371	0.389	0.369	0.395	0.380	0.400	0.376	0.395	0.381	0.396	0.395	0.425	0.477	0.476
	336	0.400±0.002	0.417±0.001	0.421	0.433	**0.382**	**0.409**	0.420	0.432	0.428	0.432	0.440	0.451	0.426	0.438	0.462	0.467	0.594	0.541
	720	**0.416±0.001**	0.437±0.002	0.453	0.458	**0.415**	**0.434**	0.430	0.446	0.427	0.445	0.436	0.453	0.431	0.446	0.721	0.604	0.831	0.657
	avg	**0.362±0.001**	**0.390±0.001**	0.383	0.404	0.365	0.393	0.378	0.404	0.383	0.407	0.385	0.410	0.383	0.408	0.472	0.465	0.559	0.515
ETTm1	96	**0.304±0.001**	**0.343±0.001**	0.319	0.360	0.326	0.361	0.318	0.358	0.334	0.368	0.329	0.365	0.355	0.375	0.335	0.372	0.345	0.372
	192	**0.354±0.001**	**0.373±0.001**	0.360	0.381	0.363	0.380	0.364	0.383	0.377	0.391	0.380	0.394	0.392	0.393	0.388	0.401	0.380	0.389
	336	**0.379±0.001**	**0.393±0.001**	0.389	0.403	0.395	0.403	0.396	0.406	0.426	0.420	0.400	0.410	0.424	0.414	0.421	0.426	0.413	0.413
	720	**0.435±0.001**	**0.428±0.001**	0.447	0.441	0.453	0.438	0.456	0.444	0.491	0.459	0.475	0.453	0.487	0.449	0.486	0.465	0.474	0.453
	avg	**0.368±0.001**	**0.384±0.001**	0.379	0.396	0.384	0.395	0.384	0.398	0.407	0.410	0.396	0.406	0.415	0.408	0.408	0.416	0.403	0.407
ETTm2	96	**0.163±0.001**	**0.242±0.001**	**0.163**	0.246	0.177	0.259	0.174	0.257	0.180	0.264	0.184	0.264	0.183	0.266	0.189	0.277	0.193	0.292
	192	**0.226±0.001**	**0.285±0.001**	0.229	0.290	0.243	0.301	0.240	0.300	0.250	0.309	0.246	0.306	0.247	0.305	0.258	0.326	0.284	0.362
	336	0.286±0.001	**0.324±0.001**	**0.284**	0.327	0.302	0.340	0.297	0.339	0.311	0.348	0.308	0.346	0.307	0.342	0.343	0.390	0.369	0.427
	720	**0.383±0.001**	**0.381±0.001**	0.389	0.391	0.397	0.396	0.392	0.393	0.412	0.407	0.409	0.402	0.407	0.399	0.495	0.480	0.554	0.522
	avg	**0.265±0.001**	**0.308±0.001**	0.266	0.314	0.279	0.324	0.276	0.322	0.288	0.332	0.287	0.330	0.286	0.328	0.321	0.368	0.350	0.401
Weather	96	**0.150±0.001**	**0.190±0.001**	0.158	0.203	0.163	0.207	0.162	0.207	0.174	0.214	0.176	0.217	0.166	0.213	0.174	0.208	0.196	0.255
	192	**0.200±0.001**	**0.238±0.001**	0.207	0.247	0.211	0.251	0.210	0.250	0.221	0.254	0.221	0.256	0.213	0.254	0.219	0.250	0.237	0.296
	336	**0.259±0.001**	**0.282±0.001**	0.262	0.289	0.267	0.292	0.265	0.290	0.278	0.296	0.275	0.296	0.269	0.294	0.273	0.290	0.283	0.335
	720	0.339±0.001	0.336±0.001	0.344	0.344	0.343	0.341	0.342	0.340	0.358	0.347	0.352	0.346	0.346	0.343	**0.334**	**0.332**	0.345	0.381
	avg	**0.237±0.001**	**0.262±0.001**	0.243	0.271	0.246	0.272	0.245	0.272	0.258	0.278	0.256	0.279	0.249	0.276	0.250	0.270	0.265	0.317
Electricity	96	**0.136±0.001**	0.231±0.001	**0.136**	**0.229**	0.147	0.241	0.147	0.245	0.148	0.240	0.164	0.251	0.200	0.278	0.176	0.258	0.197	0.282
	192	0.153±0.001	0.246±0.001	**0.152**	**0.244**	0.165	0.258	0.160	0.250	0.162	0.253	0.173	0.262	0.200	0.280	0.175	0.262	0.196	0.285
	336	**0.170±0.001**	**0.264±0.001**	**0.170**	**0.264**	0.177	0.273	0.173	0.267	0.178	0.269	0.190	0.279	0.214	0.295	0.185	0.278	0.209	0.301
	720	**0.208±0.001**	**0.297±0.001**	0.212	0.299	0.213	0.304	0.210	0.309	0.225	0.317	0.230	0.313	0.255	0.327	0.220	0.315	0.245	0.333
	avg	**0.167±0.001**	0.260±0.001	0.168	**0.259**	0.175	0.269	0.173	0.268	0.178	0.270	0.189	0.276	0.217	0.295	0.189	0.278	0.212	0.300
Traffic	96	0.432±0.002	0.279±0.001	0.458	0.296	0.406	0.277	0.430	0.294	**0.395**	**0.268**	0.427	0.272	0.651	0.391	0.593	0.378	0.650	0.396
	192	0.442±0.001	0.289±0.001	0.457	0.294	0.426	0.290	0.452	0.307	**0.417**	**0.276**	0.454	0.289	0.602	0.363	0.595	0.377	0.598	0.370
	336	0.456±0.002	0.295±0.002	0.470	0.299	**0.432**	**0.281**	0.470	0.316	0.433	0.283	0.450	0.282	0.609	0.366	0.609	0.385	0.605	0.373
	720	0.487±0.003	0.311±0.001	0.502	0.314	**0.463**	**0.300**	0.498	0.323	0.467	0.302	0.484	0.301	0.647	0.385	0.673	0.418	0.645	0.394
	avg	0.454±0.002	0.293±0.001	0.472	0.301	0.431	0.287	0.463	0.310	**0.428**	**0.282**	0.454	0.286	0.627	0.376	0.618	0.390	0.625	0.383

Ablation study

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Table 11: Ablation study results of key modules. The best results are in **bold**.

Dataset		ETTh1		ETTm2		Weather		Electricity	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TimeEmb	96	**0.366**	**0.387**	**0.163**	**0.242**	0.150	**0.190**	**0.136**	**0.231**
	192	**0.417**	**0.416**	**0.226**	**0.285**	0.200	**0.238**	**0.153**	**0.246**
	336	0.457	**0.436**	**0.286**	**0.324**	0.259	**0.282**	**0.170**	**0.264**
	720	**0.459**	**0.460**	**0.383**	**0.381**	0.339	**0.336**	**0.208**	**0.297**
	avg	**0.425**	**0.425**	**0.265**	**0.308**	0.237	**0.262**	**0.167**	**0.260**
Random	96	0.407	0.415	0.240	0.318	0.197	0.237	0.242	0.344
	192	0.453	0.443	0.299	0.351	0.249	0.279	0.251	0.354
	336	0.499	0.471	0.361	0.386	0.314	0.321	0.291	0.393
	720	0.550	0.524	0.452	0.434	0.422	0.384	0.370	0.455
	avg	0.477	0.463	0.338	0.372	0.296	0.305	0.289	0.387
Zero	96	0.405	0.414	0.240	0.317	0.198	0.237	0.239	0.342
	192	0.452	0.442	0.299	0.351	0.248	0.279	0.248	0.352
	336	0.497	0.470	0.360	0.385	0.314	0.321	0.289	0.391
	720	0.547	0.523	0.450	0.433	0.421	0.383	0.367	0.453
	avg	0.475	0.462	0.337	0.372	0.295	0.305	0.286	0.385
Mean	96	0.397	0.410	0.204	0.289	0.175	0.221	0.241	0.339
	192	0.444	0.439	0.266	0.327	0.220	0.261	0.250	0.349
	336	0.488	0.464	0.328	0.364	0.267	0.296	0.291	0.387
	720	0.537	0.517	0.427	0.418	0.346	0.348	0.374	0.453
	avg	0.467	0.458	0.306	0.350	0.252	0.282	0.289	0.382
w/o. $\boldsymbol{X}_s$	96	0.376	0.391	0.173	0.252	0.182	0.221	0.178	0.259
	192	0.431	0.422	0.237	0.294	0.228	0.259	0.184	0.266
	336	0.470	0.441	0.295	0.332	0.282	0.299	0.200	0.282
	720	0.487	0.471	0.391	0.389	0.356	0.347	0.241	0.316
	avg	0.441	0.431	0.274	0.317	0.262	0.282	0.201	0.281
w/o. $\mathcal{H}_\omega$	96	**0.366**	0.391	0.164	0.244	0.151	0.192	0.137	0.233
	192	**0.417**	0.421	0.229	0.287	0.201	0.239	0.155	0.249
	336	**0.451**	0.441	**0.286**	**0.324**	0.259	0.283	0.172	0.269
	720	0.492	0.474	0.385	0.382	0.341	0.337	0.210	0.299
	avg	0.432	0.432	0.266	0.309	0.238	0.263	0.169	0.263
w/o. RevIN	96	0.383	0.403	0.193	0.286	**0.147**	0.191	0.137	0.235
	192	0.450	0.449	0.283	0.352	**0.194**	**0.238**	0.154	0.252
	336	0.485	0.463	0.427	0.449	**0.248**	0.290	0.171	0.271
	720	0.517	0.511	0.516	0.482	**0.326**	0.346	0.213	0.306
	avg	0.459	0.457	0.355	0.392	**0.229**	0.266	0.169	0.266

Framework Performance

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Table 8: We equip Fredformer and CycleNet with TimeEmb on ETTh2. It brings stable performance(MSE) gains with trivial extra training costs.

Horizon	96	192	336	720
Fredformer	0.293	0.371	0.382	0.415
+TimeEmb	**0.289**	**0.358**	**0.360**	**0.387**
Impr.	1.4%	3.5%	5.8%	6.7%
former param	32820131	33465731	9894911	13656959
current param	32830860	33476460	9905640	13667688
extra param	**10729 (0.03%-0.11%)**			
CycleNet	0.285	0.373	0.421	0.453
+TimeEmb	**0.277**	**0.351**	**0.399**	**0.415**
Impr.	2.8%	5.9%	5.2%	8.4%
former param	99080	148328	222200	419192
current param	109809	159057	232929	429921
extra param	**10729 (2.56%-10.83%)**			